India's Agricultural Crop Production Analysis

TEAM ID	SWTID-2026-4021
PROJECT TITLE	India's Agricultural Crop Production Analysis
MAX MARKS	2 MARKS

5.2 – Sprint Delivery Plan

The Sprint Delivery Plan for the Agricultural Crop Production Analysis System follows an Agile methodology, where the project is divided into short, iterative sprints. Each sprint delivers a working increment of the system, ensuring continuous improvement, faster feedback, and better alignment with user needs such as farmers, agricultural officers, and policymakers.

Sprint Duration

- Each sprint: 2 weeks
- Total sprints: 5–6 sprints (approx.)
- Methodology: Agile / Scrum

Sprint 1: Requirement Analysis & System Setup

Goals

Establish project foundation and finalize requirements.

Tasks

- Gather functional and non-functional requirements.
- Identify stakeholders (farmers, analysts, government).
- Define system scope and objectives.
- Set up development environment and tools.
- Design initial project architecture.

Deliverables

- Requirement Specification Document
- Project roadmap
- Initial system architecture

Sprint 2: UI/UX Design & Database Setup

Goals

Design system interface and prepare data storage structure.

Tasks

- Create wireframes and dashboard layouts.
- Design user interface for web/mobile application.
- Develop database schema (crop, weather, market data).
- Set up MySQL/PostgreSQL database.
- Define API structure.

Deliverables

- UI mockups
- Database design
- Initial working database setup

Sprint 3: Backend Development & API Integration

Goals

Build core backend functionality.

Tasks

- Develop backend using Python (Django/Flask).
- Create REST APIs for data handling.
- Implement data collection modules.
- Integrate external APIs (weather, market data).
- Connect backend with database.

Deliverables

- Functional backend system
- Working APIs
- Data integration framework

Sprint 4: Analytics & Machine Learning Integration

Goals

Enable crop analysis and prediction features.

Tasks

- Data preprocessing and cleaning.
- Develop machine learning models for yield prediction.
- Implement crop trend analysis algorithms.
- Add forecasting modules (weather & production).
- Validate model accuracy.

Deliverables

- ML prediction models
- Analytics engine
- Crop yield forecasting system

Sprint 5: Frontend Integration & Dashboard Development

Goals

Build interactive user interface with analytics visualization.

Tasks

- Connect frontend with backend APIs.
- Develop dashboards for crop insights.
- Create charts, graphs, and reports.
- Implement user-friendly navigation.
- Enable mobile responsiveness.

Deliverables

- Fully functional frontend
- Interactive dashboards
- Data visualization system

Sprint 6: Testing, Deployment & Optimization

Goals

Ensure system stability and deploy final product.

Tasks

- Perform unit and integration testing.
- Conduct user acceptance testing (UAT).
- Fix bugs and optimize performance.
- Deploy system on cloud (AWS/Azure/GCP).
- Prepare documentation and user training.

Deliverables

- Fully deployed system
- Final project documentation
- Test reports and optimized application

Key Agile Practices

- Daily stand-up meetings
- Sprint review sessions
- Continuous feedback from stakeholders
- Iterative improvements after each sprint

Expected Outcomes

- Faster development with incremental delivery.
- Early detection and correction of issues.
- Better alignment with farmer and stakeholder needs.
- Improved system quality and usability.
- Scalable and adaptive agricultural analytics platform.

Conclusion

The Sprint Delivery Plan ensures structured, iterative, and efficient development of the Agricultural Crop Production Analysis System. By delivering features in phases, the project achieves continuous improvement, higher reliability, and effective support for agricultural decision-making in India.